The Right Triangle Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: PYTHAG, SPECIAL, TRIG and APPLY (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** PYTHAG
- ☐ **A2** PYTHAG
- ☐ **A3** SPECIAL
- ☐ **A4** SPECIAL
- ☐ **A5** TRIG
- ☐ **A6** APPLY

Entrées

10 pts each · choose 4

- ☐ **E1** PYTHAG
- ☐ **E2** PYTHAG
- ☐ **E3** APPLY
- ☐ **E4** SPECIAL
- ☐ **E5** TRIG
- ☐ **E6** TRIG

Chef's Specials

20 pts each · choose 2

- ☐ **S1** PYTHAG
- ☐ **S2** SPECIAL
- ☐ **S3** TRIG
- ☐ **S4** APPLY

My skill check (circle each skill you have covered): PYTHAG · SPECIAL · TRIG · APPLY

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 PYTHAG ☐ My pick · 5 pts

A right triangle has legs 9 and 12. Find the length of the hypotenuse.

A2 PYTHAG ☐ My pick · 5 pts

Could 7, 24, and 25 be the side lengths of a right triangle? Show your work.

A3 SPECIAL ☐ My pick · 5 pts

A 45°-45°-90° triangle has legs of length 5. Find the hypotenuse in simplest radical form.

A4 SPECIAL ☐ My pick · 5 pts

A 30°-60°-90° triangle has a shorter leg of 4. Find the hypotenuse and the longer leg.

A5 TRIG ☐ My pick · 5 pts

In $\triangle ABC$, $\angle C = 90°$, $AB = 10$, $BC = 6$, and $AC = 8$. Write sin A, cos A, and tan A as fractions in lowest terms.

A6 APPLY ☐ My pick · 5 pts

A 13 ft ladder leans against a wall and reaches 12 ft up the wall. How far is the base of the ladder from the wall?

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 PYTHAG ☐ My pick · 10 pts

A rectangle has a diagonal of 17 and a width of 8. Find its length and its perimeter.

E2 PYTHAG ☐ My pick · 10 pts

Which of these could be the sides of a right triangle? Show the test for each. (a) 10, 24, 26 (b) 9, 40, 42 (c) 12, 35, 37

E3 APPLY ☐ My pick · 10 pts

A wheelchair ramp rises 1.5 ft over a horizontal run of 18 ft. Find the angle of the ramp with the ground and the length of the ramp surface, each to the nearest tenth.

E4 SPECIAL ☐ My pick · 10 pts

An equilateral triangle has side length 10. Draw an altitude and use a special right triangle to find the altitude and the area of the triangle, in simplest radical form.

E5 TRIG ☐ My pick · 10 pts

In $\triangle PQR$, $\angle Q = 90°$, $\angle P = 35°$, and $PR = 20$. Find QR and PQ to the nearest tenth. ($\sin 35° \approx 0.5736$, $\cos 35° \approx 0.8192$)

E6 TRIG ☐ My pick · 10 pts

In right $\triangle ABC$ ($\angle C = 90°$) the side opposite $\angle A$ is 5 and the hypotenuse is 13. Kai writes $\tan A = \frac{5}{13}$, so $A = \tan^{-1}(\frac{5}{13}) \approx 21°$. Find his mistake and give the correct angle to the nearest degree.

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

Name: _____

S1 PYTHAG ☐ My pick · 20 pts

In △ABC, D is on side BC, AD ⊥ BC, AB = 13, AC = 20, and AD = 12. Find BD, DC, BC, and the perimeter of △ABC. Then use the converse of the Pythagorean theorem to decide whether △ABC is a right triangle.

S2 SPECIAL ☐ My pick · 20 pts

Rectangle ABCD has diagonal AC = 20 and ∠CAB = 30°. (a) Explain why △ABC is a 30°-60°-90° triangle. (b) Find AB and BC in simplest radical form. (c) Find the area and perimeter of the rectangle.

S3 TRIG ☐ My pick · 20 pts

In right △ABC, ∠C = 90°, $\sin A = \frac{7}{25}$, and AB = 50. (a) Find BC. (b) Find AC. (c) Find cos A and tan A. (d) Check your sides with the Pythagorean theorem.

S4 APPLY ☐ My pick · 20 pts

A boat is 120 m from the base of a sea cliff. The angle of elevation to the top of the cliff is 25°, and to the top of a lighthouse at the cliff’s edge it is 32°. (tan 25° ≈ 0.4663, tan 32° ≈ 0.6249) (a) Find the cliff height and the lighthouse top height above sea level, to the nearest tenth. (b) Find the lighthouse height. (c) From the lighthouse top, what is the angle of depression to the boat, and why?